An Innovative Low-Altitude Dual-Source Remote Sensing Platform for Urban Thermal Environment Observation: Integration, Tests, Optimization, and Assessment

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***Abstract*—The urban thermal environment significantly influences regional climate change, urban resilience, and sustainability. Although traditional remote sensing technologies are common tools for observing urban thermal environments, it has limitations, including low spatial resolution, long revisit cycles, and cloud cover. In responding to these limitations, a low-altitude dual-source remote sensing platform (LADSRSP) was researched and developed in this study through a series of tests. First, a thermal infrared sensor and a hyperspectral sensor were simultaneously equipped on a multirotor drone via a customized three-axis gimbal in LADSRSP. Furthermore, LADSRSP was optimized based on the stability test results and was then confirmed to have the ability to collect data with high spatial resolution and quality through the data acquisition tests. Finally, the performance of LADSRSP was assessed by data integrity, classification accuracy, and temperature analysis. The image classification results demonstrate high accuracy with an overall classification accuracy of 97.92% and a Kappa coefficient of 0.97 from collected data quality analysis tests. The temperature values of each urban underlying surface were then accurately extracted to facilitate further statistical analysis. Overall, the LADSRSP has been proven to be a feasible, efficient, and accurate tool for urban thermal environment research.**

***Index Terms*—Dual-source remote sensing, low altitude, platform integration, technological innovation, urban thermal environment.**

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I. Introduction

THE current acceleration of urbanization has brought serious challenges to urban thermal environment issues. Rapid economic development and urbanization have led to the continuous expansion of urban spaces and a substantial migration of rural residents to cities [1], [2]. Data from 2023 indicate that 64.6% of China's population resided in urban areas (databank.worldbank.org), and this proportion is expected to continue growing [3]. Concurrently, the heat radiation stored by impervious surfaces in cities, combined with carbon emissions from human activities and industrial production, has exacerbated the deterioration of the thermal environment [4], [5], [6], [7]. One of the primary manifestations of deterioration is the urban heat island (UHI) effect, which is intimately related to various environmental and public health issues [8], [9], [10], [11], [12]. Notably, heat stroke incidents caused by high temperatures have occurred frequently in recent years, with the mortality rate of hospitalized patients ranging from 14% to 60% [13], [14], [15], [16]. Consequently, in-depth research on urban thermal environments is essential to advancing the understanding of improving environmental livability, ensuring public health, reducing the UHI effect, and promoting sustainable urban development [17], [18], [19], [20].

Existing environmental monitoring methods have limitations in urban thermal environment observation. Traditional ground measurement stations can accurately collect environmental monitoring data, such as air temperature, humidity, wind speed, and pollutant concentrations [21], [22]. Satellite remote sensing technologies enable the acquisition of large-scale thermal infrared and multispectral data, which are instrumental in analyzing UHI effects and studying strategies to improve the thermal environment [23], [24], [25]. It should be noted that using land surface temperature (LST) as a single indicator for UHI is limited since the UHI effect could also be affected by air temperature, ventilation conditions, and urban structure [26], [27], [28]. However, these monitoring methods are not universally applicable. Ground measurement stations,

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constrained by their spatial coverage, struggle to accurately depict the distribution characteristics of extensive urban thermal environments with a limited number of installations [29], [30]. Although satellite remote sensing, with its broad spatial coverage and global reach, has become the primary means for urban thermal environment observation, its longer revisit cycles and susceptibility to cloud cover may not be conducive to high-precision research [30], [31], [32].

In recent years, low-altitude remote sensing technologies have been widely applied to study the urban thermal environment due to their high spatial resolution and spatiotemporal flexibility. For instance, Kim et al. [32] have conducted a comparative analysis of ground temperature images obtained from both low-altitude and satellite remote sensing. Their study affirms the reliability of low-altitude remote sensing for precise LST measurements, demonstrating superior accuracy over satellite technologies. Furthermore, Di Felice et al. [33] have demonstrated that a drone with a thermal infrared camera is effective for rapid and accurate thermal analysis in small to medium areas.

Although low-altitude single-source remote sensing technologies effectively observe urban thermal environments and assess data accuracy, they are insufficient to capture the full complexity of urban environments, particularly in the presence of diverse vegetation and various artificial surfaces [34], [35], [36]. Consequently, multirotor drones equipped with RGB and thermal infrared cameras were used to collect dual-source remote sensing data, which is an effective strategy for characterizing the 3-D thermal properties of forest canopies [37]. An RGB image, which encompasses only red, green, and blue bands, presents significant limitations in plant status analysis and object classification accuracy. To address these limitations, this article introduces an innovative low-altitude dual-source remote sensing platform (LADSRSP) equipped with hyperspectral and thermal infrared sensors for urban thermal environment observation, thereby providing a more comprehensive and accurate data acquisition tool.

Currently, there is a lack of comprehensive research on the integration and optimization of low-altitude nonsingle-source remote sensing platforms and specifically designed for urban thermal environment studies. A significant challenge in the application of LADSRSP for urban thermal environment research is ensuring stable performance and high-precision data acquisition capabilities. In this regard, complete integration schemes and optimization strategies should be provided, along with a series of performance tests to assess the applicability and advantages of using LADSRSP to study urban environments.

The overall objective of the present study is to develop and optimize the LADSRSP to synchronously observe urban thermal environments with high spatial and spectral resolution. Accordingly, this study aims to: 1) elucidate the integration scheme of LADSRSP by introducing its system components, equipment selection, operational mechanism, and connection methods; 2) evaluate the functionality and performance of LADSRSP by conducting stability tests, data acquisition tests, and collected data quality analysis tests; 3) optimize the integration scheme of LADSRSP based on the test results; and 4) assess the aerial photography performance of LADSRSP based on data spatial resolution, classification accuracy, data integrity, and temperature analysis in low-altitude thermal environment observations.

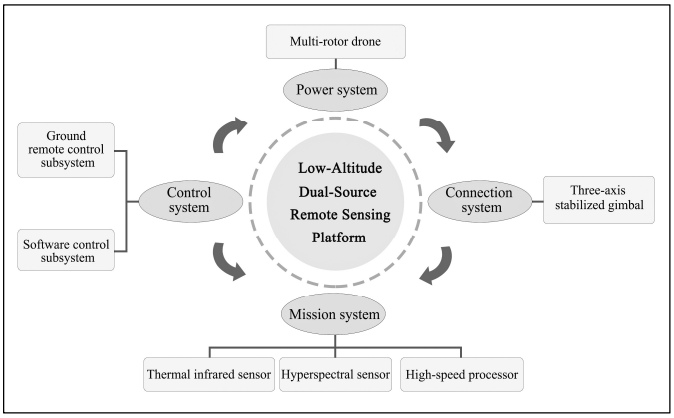


Fig. 1. System components of LADSRSP and the equipment composition of each system.

II. Platform Overview

A. System Components

The LADSRSP is composed of a power system, a mission system, a connection system, and a control system, as illustrated in Fig. 1. The power system consists of a multirotor drone that is tasked with carrying out low-altitude aerial photography operations. The mission system consists of a thermal infrared sensor, a hyperspectral sensor, and a high-speed processor. The connection system consists of a three-axis stabilized gimbal to connect the power system and the mission system while controlling the orientation of the mission system during operation. The control system includes a ground remote control subsystem and a software control subsystem, which are used to set the operating parameters of the power system and mission system, respectively.

B. Equipment Selection Scheme

The DJI M300 RTK multirotor drone was selected as the flight equipment for the power system of the LADSRSP. This drone has a payload of 2.7 kg, a wind resistance of 12 m/s, and a maximum hovering accuracy of ± 0.1 m. Third-party payloads (e.g., camera, sensor, and searchlight) developed based on the SkyPort V2 standard interface can be connected to this drone through the quick-release interface on the front of the fuselage.

The FLIR VUE PRO R thermal infrared sensor was selected as the equipment for measuring surface radiation temperature in the mission system of the LADSRSP. This sensor detects thermal infrared radiation in the wavelength range of 7.5–13.5 μm at a field of view of $32^\circ \times 24^\circ$. The image capture frequency of this sensor is configured to range from 1 to 0.017 Hz (one frame every 60 s). The thermal infrared data are automatically collected at the predetermined frequency and stored in the plug-in memory card. The OPTOSKY ATH1010W push-broom hyperspectral sensor was selected

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TABLE I
SPECIFICATIONS AND COMPETENCY OF THE EQUIPMENT OF LADSRSP

Classification	Model	Specification	Competency	Appearance
Drone	DJI M300 RTK	Dimensions: 430 mm × 420 mm × 430 mm (Folding, including paddles) Weight: 6.3 kg (Includes dual batteries) Max speed: 23 m/s Flight time: Approximately 55 minutes (Dual batteries) Max payload：2.7 kg	High payload capacity, strong wind resistance, and vertical takeoff and landing capability.	
Sensor	FLIR VUE PRO R	Dimensions: 57.4 mm × 44.5 mm × 44.5 mm Weight: 92.14 g Resolution: 640 ×512 pixels Spectral range: 7.5 to 13.5 μm Scene range: −25 to 135 ℃ (High gain) Accuracy: ±5 ℃ or 5% of the reading Focal length: 19 mm Field of view: 32° × 24°	Compact, light weight, high sensitivity, and high accuracy.	
	OPTOSKY ATH1010 W	Dimensions: 239.7 mm × 68 mm × 84.9 mm Weight: 680 g Spectral resolution：1.47 nm Spectral range: 400 to 1000 nm Sensor resolution: 2048 × 2048 pixels Focal length: 25 mm Field of view: 25.4°	Light weight, high spectral resolution and wide spectral range.	
Processor	Customized	Dimensions: 66 mm × 66 mm × 82 mm Weight: 200 g Fuselage interface: Type-A、Type-C、HDMI	Compact, light weight, real-time processing and storage capabilities.	
Gimbal	Customized	Dimensions:129 mm × 112 mm × 230 mm Angle jitter：0.02° Heading control rotation range：±290° Pitch control rotation range：-40° to 90°	Simultaneous integration of dual-source sensors, accurate angle control, avoidance of sensor sway.	

as the equipment for collecting spectral information in the mission system of the LADSRSP. This sensor captures spectral data in the wavelength range of 400–1000 nm at a spectral resolution of 1.47 nm, with an instantaneous field of view (IFOV) of 0.88 mrad.

The customized high-speed processor was used to process and store the hyperspectral data. The signal of each band collected by the hyperspectral sensor is processed by the high-speed processor in real time and converted into an electrical signal representing the spectral intensity. Besides, the acquisition parameters (e.g., exposure time and acquisition frame rate) for the hyperspectral sensor are set in the high-speed processor.

A three-axis stabilized gimbal was customized as the connecting equipment in the connection system of the LADSRSP. The gimbal includes a quick-release interface, gimbal arms, three-axis motors, and a mounting compartment. The motors provide a heading control rotation range of ±290° and a pitch control rotation range of −40° to 90°, leading to precise control of the orientation. This extensive control range ensures that the two sensors remain synchronized and fixed at any angle within the specified rotation ranges. The specifications for the equipment described above are provided in detail, as shown in Table I.

C. Connection Method and Operational Mechanism

In the hardware connection, the DJI M300 RTK multirotor drone interfaces with the three-axis stabilized gimbal through the DJI Gimbal Connect 2.0 (DGC 2.0). The hyperspectral sensor is firmly integrated into the gimbal through the mounting compartment, while the thermal infrared sensor is mounted to the gimbal using a custom 3-D-printed bracket. The high-speed processor is positioned atop the hyperspectral sensor and supplies electrical power to the two sensors via an external lithium battery. The hardware connections and position between the drone, gimbal, sensors, and processor are shown in Fig. 2. In the software connection, the thermal

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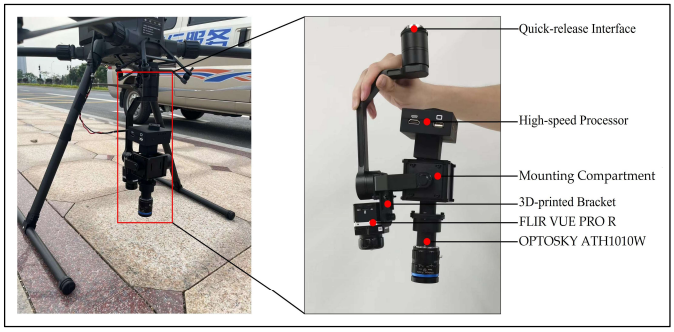


Fig. 2. (Left) Appearance of the LADSRSP and (right) illustration of mission system hardware connections.

infrared sensor is wirelessly connected to the software control subsystem through Bluetooth pairing protocol. Signal synchronization between the multirotor drone and ground remote control subsystem is facilitated by the DJI Plot 2 software (DJI, Shenzhen, Guangdong, China).

The operations of the LADSRSP for urban thermal environment observation are mainly divided into two primary phases: route planning and data acquisition. During route planning, the observation area is first selected on the online map in the ground remote control subsystem. The flight route is then defined based on the specified parameters (e.g., altitude, speed, and lateral overlap ratio). The acquisition parameters (e.g., altitude, speed, and frame rate) for the hyperspectral sensor are set to match the flight route parameters within the high-speed processor. An appropriate exposure time of 5000 μs is typically configured for the hyperspectral sensor under clear weather conditions.

In the data acquisition phase, the planned route is uploaded to the multirotor drone, after which the route task is initiated from the ground remote control subsystem. The drone automatically follows the designated flight route after being raised to a certain altitude under manual operation, collecting data based on the preconfigured parameters. During this process, the flight status of the platform could be checked through the real-time monitoring interface of the ground remote control subsystem. Collected thermal infrared and hyperspectral data are stored in the built-in memory card (512 GB, Samsung, SD card). The platform is operated manually by the ground remote control subsystem to return and land after data collection.

III. Test Methods

A. Observation Area and Tests

The general strategy of the test program for the LADSRSP includes stability test, data acquisition test, and collected data quality analysis test. Nearly 100 relevant tests were carried out in this study over 14 months. Several representative tests from each stage were selected to illustrate the methods and results, providing a comprehensive understanding of the test process and its impact on LADSRSP. The test area is located close to the intersection of Sanshe Road and Fengqi Road in Jimei District, Xiamen City, Fujian Province, as shown in Fig. 3.

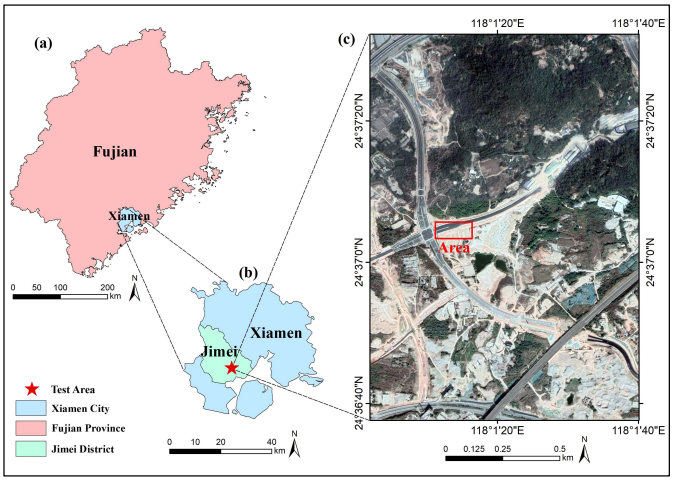


Fig. 3. Map of test area locations. (a) Xiamen City within Fujian Province, China. (b) Test area in Jimei District, Xiamen City. (c) Satellite image of the test area at the intersection of Sanshe Road and Fengqi Road.

Stability tests were conducted both on June 15 and October 17, 2023, at an altitude of 100 m and a flight speed of 4 m/s, to evaluate flight performance (e.g., significant deflection, torsion, and motor noise) through aerial observation and image analysis. The data of the inertial measurement unit (IMU) integrated within the hyperspectral sensor were collected during the tests to capture the angular variations of the sensor along the heading, roll, and pitch axes. Among them, the angle variations of the roll and pitch axes have a significant impact on the imaging quality [38].

Data acquisition tests were conducted both on November 14 (from 15:24 P.M. to 15:37 P.M.) and December 28 (from 10:00 A.M. to 10:09 A.M.), 2023, at an altitude of 100 m. The flight speeds were set at 8 and 4 m/s to evaluate the data collection capabilities of the platform at different flight speeds. The tests were carried out under clear weather conditions with few clouds. The solar elevation angle for the test on November 14 ranged between 22.97° and 20.51°, whereas the solar elevation angle for the test on December 28 ranged between 33.35° and 34.48°. The integrity of the hyperspectral data files (including spectral data, timestamps data, IMU data, and configuration data) and the amount of thermal infrared images were used as the indicators of sensor performance in the data acquisition tests. The distortion or compression in the image was detected by visual interpretation during the image quality inspection.

Collected data quality analysis tests were conducted both on January 10 and May 15, 2024. In January test, the flight was performed at a speed of 1 m/s and an altitude of 100 m. In May test, the flight parameters were adjusted to a speed of 6 m/s and an altitude of 113 m. Thermal infrared images acquired at 1 m/s were used to verify the integrity of hyperspectral images captured in push-broom mode. Referring to the emissivity setting for urban underlying surface types from the research by Ortiz-Sanz et al. [39], the emissivity parameter in the thermal infrared sensor was set at 0.95 as a general estimate, to provide a reasonable approximation for the emissivity of several different underlying surface types.

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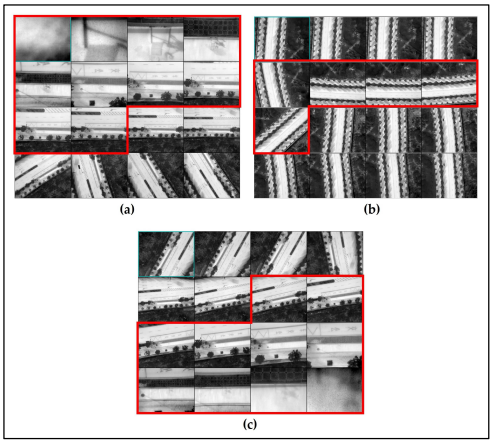


Fig. 4. Thermal infrared images of transitional stages outlined with red wireframe. (a) Takeoff stage: the platform takes off from the ground to the set height. (b) Turning stage: the platform turns at the end of the route. (c) Landing stage: the platform lands from the set height to the ground.

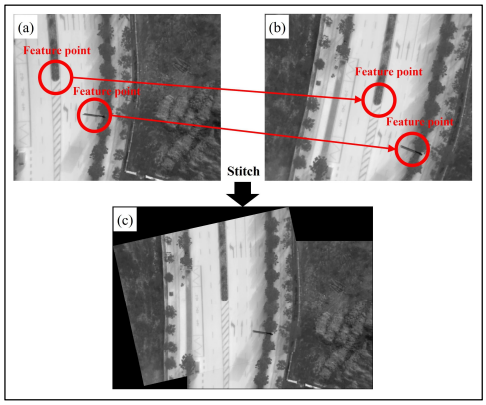


Fig. 5. Adjacent thermal infrared images stitching process. (a) Identified feature points in a thermal infrared image. (b) Corresponding feature points in the adjacent thermal infrared image. (c) Mosaic of the stitched result.

Image stitching and registration, hyperspectral classification, classification accuracy assessment, LST calculation, and analysis were carried out in the data processing.

B. Image Stitching and Registration

Image stitching process of the thermal infrared images was conducted in Pix4Dmapper (version 4.5.6, Pix4D, Swiss) and mainly divided into two steps. First, the images were imported into FLIR Tools (version 5.13, Teledyne FLIR, Wilsonville, OR, USA) to identify and filter out the images of transitional stages, such as takeoff, landing, and turning stages. The filtering of images at these stages by visual interpretation is shown in Fig. 4. Second, the correspondence was established among images through the identification of feature points (e.g., corner points, edge points, or points with significant features) to obtain the mosaic, as shown in Fig. 5.

The georeferencing process for the thermal infrared mosaic of the whole observed area was conducted in ArcGIS Desktop (version 10.8.1, Esri, Redlands, CA, USA) and was mainly divided into three steps. First, the mosaic was imported into ArcGIS for georeferencing. Second, a remote sensing image

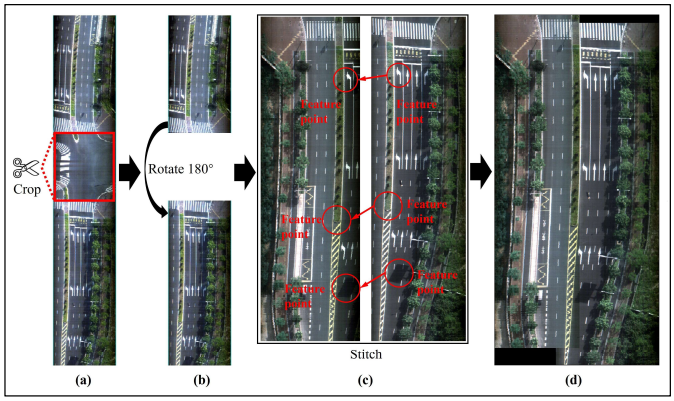


Fig. 6. Hyperspectral image stitching process. (a) Original hyperspectral image fragment. (b) Croped hyperspectral fragments. (c) Feature points identification in adjacent hyperspectral fragments. (d) Stitched result of hyperspectral image.

with precise geographic coordinates obtained from Google Earth Pro (version 7.3.6, Google, Mountain View, CA, USA) was loaded into ArcGIS as a reference object. Third, at least three feature points were marked between the thermal infrared mosaic and the reference object. In this step, feature points with significant errors compared to others will be removed or replaced to improve georeferencing accuracy. Finally, the thermal infrared mosaic was transformed based on the feature point marking results to match the reference object, and then, geographic coordinate information was stored in the thermal infrared mosaic using georeferencing tools.

The stitching process of the hyperspectral images was conducted in HSmosaic software (OPTOSKY, Xiamen, China). First, the turning stages of the flight route were identified by observing the continuous extrusion regions through visual interpretation. Second, these stages were used as References to crop the initial images to obtain the hyperspectral fragments corresponding to each flight route. Finally, the adjacent hyperspectral fragments were stitched based on the identification of feature points to obtain the composite hyperspectral image, as illustrated in Fig. 6. The registration of the composite hyperspectral image was conducted in ArcGIS based on feature points, while the thermal infrared image with geographic coordinates was selected as the reference object.

C. Hyperspectral Image Classification and Accuracy Assessment

Maximum likelihood classifier (MLC), support vector machine (SVM), random forest (RF), and deep learning methods were common methods for image classification [40], [41], [42]. In this study, we choose SVM to classify the hyperspectral image, as it is particularly suitable for processing high-dimensional data and can effectively extract the characteristics even with relatively few training samples [43].

Sample creation and image classification were completed in ENVI (version 5.6, NV5 Geospatial, Hollywood, FL, USA). The areas of major features (e.g., road, vegetation, soil, tile, and landmark line) were extracted from the image to establish the region of interest (ROI). Finally, the created samples

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were used as a reference to classify the hyperspectral image according to the rules of SVM.

The accuracy of hyperspectral image classification was evaluated by a confusion matrix and K-fold cross validation. The confusion matrix was used to compute the overall classification accuracy and Kappa coefficient, which were then utilized to evaluate the reliability and effectiveness of each hyperspectral image classification. The K-fold cross validation is a model validation technique that involves dividing the dataset into k parts. In one iteration, one part serves as the training set, while the remaining $k-1$ parts are used for validation. In subsequent iterations, one part previously used as the validation set is designated as the training set, with the remaining parts serving as the validation set. Each part is used as the training set exactly once. After k iterations, all data parts have been used as training and validation sets for classification tests, thereby enabling a more comprehensive evaluation of classification accuracy.

In this study, the dataset was divided into three parts based on the coverage of the test data, indicating that the cross validation involved three iterations of $K1$–$K3$. For each iteration, two sets of confusion matrix calculation results were obtained for each validation. The final evaluation metrics for classification were determined by averaging the overall classification accuracy and Kappa coefficient in these iterations.

D. LST Calculation and Analysis

The LST calculation was conducted in ENVI. Each pixel in a 16-bit TIFF format thermal infrared image was represented as a digital number (DN) value, indicating the radiant intensity at the corresponding location. These DN values were stored with a scale factor of 0.04, as the thermal infrared sensor has 0.04 °C pixel sensitivity in high gain mode [44]. Afterward, a linear (1) was applied to convert the DN values registered by the thermal infrared sensor into degree Celsius (FLIR 2016)

$$T = \mathrm{DN} \times 0.04 - 273.15. \tag{1}$$

The temperature distribution of the urban underlying surfaces was analyzed based on the results of the LST calculation and the classification. Thermal infrared data and hyperspectral data were combined in this step. First, image cropping and stitching were carried out on the two types of data. Second, the complete thermal infrared and hyperspectral images of the test area were registered in ArcGIS by adding the geographic coordinate information, ensuring every pixel in both images represents the identical spot on the ground. Finally, the classification results were used to generate mask files for each class in ENVI, and the temperature data for different underlying surface types were extracted based on the matching relationship between each type of mask file and the LST.

Statistical analysis (e.g., mean, maximum, minimum, median, skewness, and kurtosis) for the temperature data of nearly 15 million pixels was then carried out in MATLAB (version R2018a, MathWorks, Natick, MA, USA) to further reveal the thermal characteristics of each surface type. Moreover, temperature distribution probability curves were plotted in MATLAB, thus enabling a visual assessment of the temperature distribution on different surfaces.

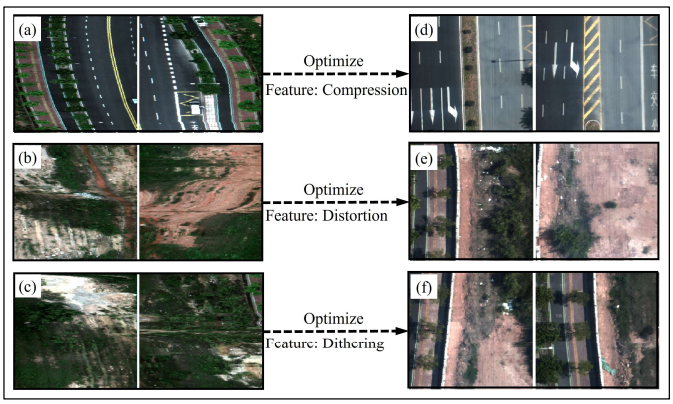


Fig. 7. Pre- and post-optimization image assessment. (a) Image compression. (b) Image distortion. (c) Image dithering. (d) Optimized results of image compression. (e) Optimized results of image distortion. (f) Optimized results of image dithering.

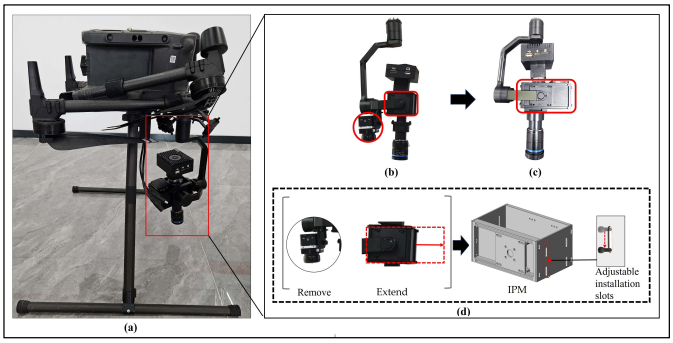


Fig. 8. Structural optimization of the sensor fixation method. (a) Overall appearance of the LADSRSP. (b) Structure before optimization. (c) Structure after optimization, with red wireframes marking the location of structural changes. (d) Design process of the IPM.

IV. Results

A. Stability and Optimization Result

Among the image quality inspection, image compression [Fig. 7(a)], image distortion [Fig. 7(b)], and image dithering [Fig. 7(c)] could be observed from the hyperspectral image fragments. To mitigate this issue, the sensor fixation method was optimized. Specifically, an integrated payload module (IPM) was designed to replace the 3-D-printed bracket and mounting compartment by extending the original design and adding adjustable installation slots, allowing fine-tuning of the vertical height of the two sensors, as depicted in Fig. 8. In the subsequent stability test, the data before and after optimization were compared and analyzed. The results indicated that the fixation method of IPM significantly improved image quality by reducing image distortion and compression, as demonstrated in Fig. 7(d)–(f).

The IMU attitude data recorded by the hyperspectral sensor after the structural optimizations are shown in Fig. 9, exhibiting the angle changes of the sensor in the roll and pitch directions over the test time. As indicated in Fig. 10, the sensor exhibited an angular change range (i.e., difference between the minimum and maximum angle values) of 2.81° on the roll axis and 4.34° on the pitch axis. The standard deviations (i.e., the degree of dispersion between each value

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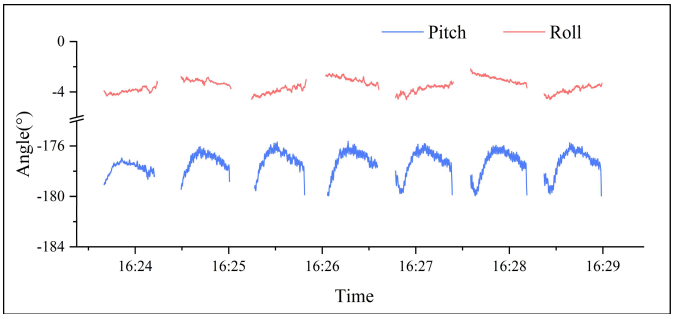


Fig. 9. Pitch and roll angle variations of hyperspectral sensors over time.

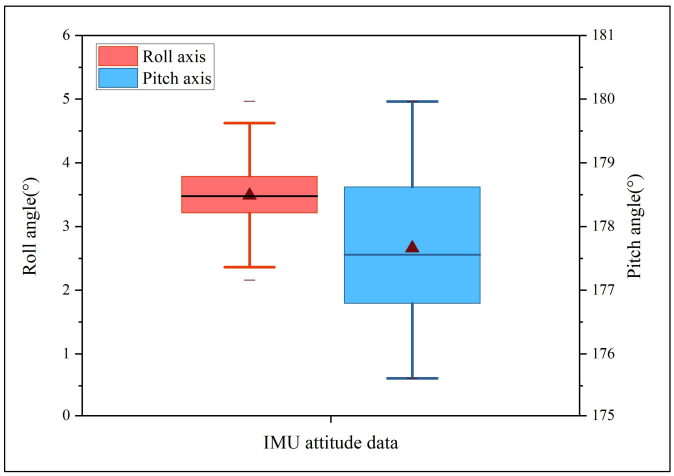


Fig. 10. Box plots of IMU attitude data from the hyperspectral sensor.

and the average value) were 0.524° for the roll axis and 0.914° for the pitch axis.

B. Results of Data Acquisition and Processing

In the test conducted on November 14, 12.5 GB of hyperspectral data and 747 thermal infrared images were collected in 12 min and 27 s at a flight speed of 8 m/s, covering a set of four parallel flight routes. In another test conducted on December 28, 16 GB of hyperspectral data and 580 thermal infrared images were collected in 9 min and 40 s at a flight speed of 4 m/s, covering a set of five parallel flight routes.

A mosaic with DN values ranging from 7427 to 8215 (Fig. 11) was created after the thermal infrared image stitching process. The stitching result of the hyperspectral image fragments (Fig. 12) was presented as a composite image of the bands at 630, 600, and 575 nm. The thermal infrared mosaic was resampled to 9.3 cm with spatial resolution during the georeferencing. The composite hyperspectral image was resampled to 9.4 cm with spatial resolution during the registration. The overlapping image area of the registered thermal infrared mosaic and composite hyperspectral image covers 124.6 × 303.8 m.

The LST calculation results (Fig. 13) indicate that the temperature ranges from 23.89 °C to 55.41 °C. Among the classification results, five types of urban underlying surfaces were extracted and shown in Fig. 14. The acreage and proportion results (Table II) indicate that the vegetation areas

TABLE II
ACREAGE AND PROPORTION OF FIVE URBAN UNDERLYING SURFACE TYPES

Underlying surface type	Acreage(m²)	Proportion(%)
Road	7075.47	18.69
Vegetation	18148.65	47.94
Soil	8941.82	23.62
Tile	2290.35	6.05
Landmark line	1400.71	3.70

(18148.65 m^2, 47.94%) covered the largest surface area, followed by soil areas (8941.82 m^2, 23.62%), road areas (7075.47 m^2, 18.69%), tile areas (2290.35 m^2, 6.05%), and landmark line areas (1400.71 m^2, 3.70%).

C. Result of Classification Accuracy and Temperature Distribution

The results of the confusion matrix (Fig. 15) for the hyperspectral image classifications indicate that the overall classification accuracy ranges from 97.11% to 98.46%, and the Kappa coefficient ranges from 0.96 to 0.98. The mean of the overall classification accuracy is 97.92%, whereas the mean of the Kappa coefficient is 0.97.

The LST of the five underlying surfaces (Fig. 16) indicates that the temperature of the road surface ranges from 25.45 °C to 55.17 °C, with an average of 48.76 °C and a median of 50.25 °C. For the vegetation, the temperature ranges from 24.53 °C to 55.13 °C, with an average of 34.02 °C and a median of 33.17 °C. For the soil, the temperature ranges from 24.29 °C to 54.53 °C, with an average of 31.47 °C and a median of 30.45 °C. For the tile, the temperature ranges from 25.37 °C to 55.05 °C, with an average of 41.72 °C and a median of 42.89 °C. For the landmark line, the temperature ranges from 25.69 °C to 55.09 °C, with an average of 46.83 °C and a median of 48.93 °C.

Moreover, the temperature distribution probability curve of each underlying surface, as depicted in Fig. 17, reveals that the soil surface exhibits the largest skewness (1.97), followed by vegetation (1.52), tile (−0.50), landmark line (−1.08), and road (−1.77). The soil surface exhibits the largest kurtosis (8.09), followed by road (6.45), vegetation (5.98), landmark line (3.18), and tile (2.40).

V. DISCUSSION

A. Assessment of Aerial Photography Performance

In the integration and tests, we designed and customized a three-axis stabilized gimbal and IPM to mount thermal infrared and hyperspectral sensors, thus maintaining the aerial stability of LADSRSP. Three-axis stabilized gimbal shows effectiveness in improving data acquisition quality in various aerial photography systems [45], [46]. Furthermore, in the flight performance testing of multirotor uncrewed aerial vehicles (UAVs), excessive and rapid changes in roll and pitch (due to an underperforming gimbal or no gimbal) of the sensors have a significant impact on the data quality [38]. Therefore, we evaluate the aerial photography performance of the LADSRSP by analyzing the angle changes of the sensors on the roll and pitch axes through IMU data.

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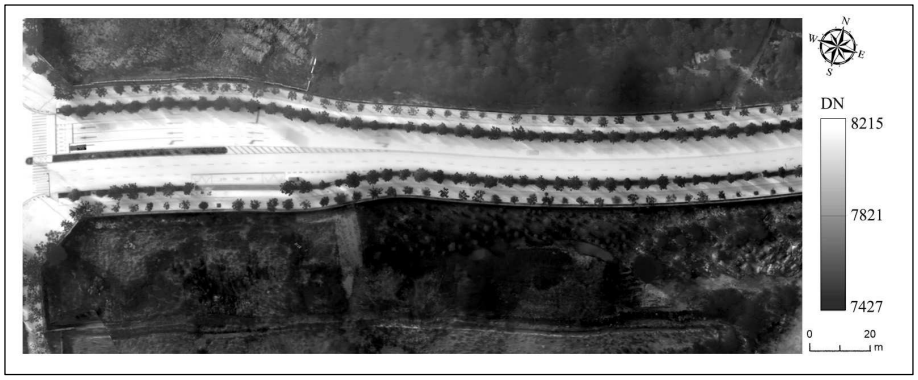


Fig. 11. Thermal infrared image stitching and georeferencing result: a mosaic representing the DN value of the underlying surface with a resampled spatial resolution of 9.3 cm.



Fig. 12. Hyperspectral image stitching and registration result: a composite hyperspectral image of the underlying surface with a resampled spatial resolution of 9.4 cm.

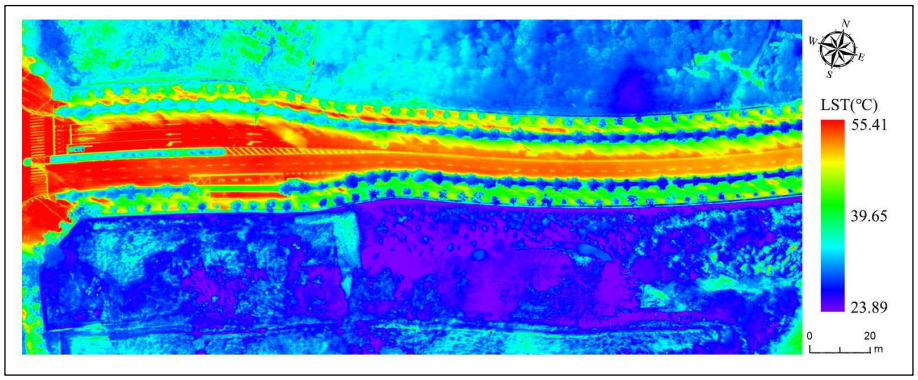


Fig. 13. LST distribution of underlying surfaces in the observation area.

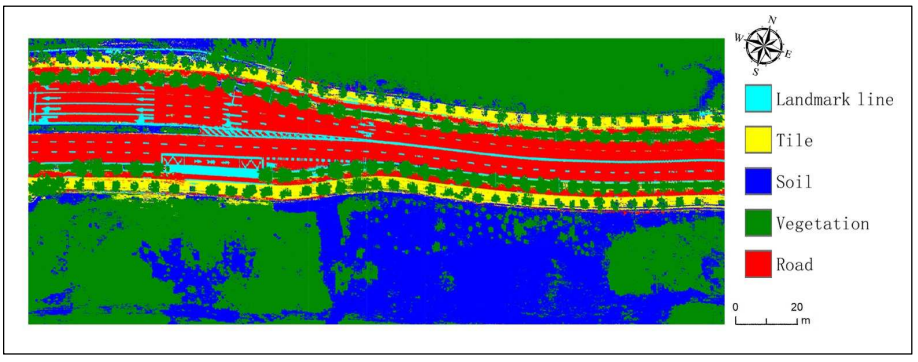


Fig. 14. Classification results of underlying surfaces in the observation area.

The result indicates that the standard deviation of the roll axis and the pitch axis are 0.524° and 0.914°, respectively. As reflecting the angular deviations of the sensors, the standard deviation result (both below 1° in the roll and pitch axes)

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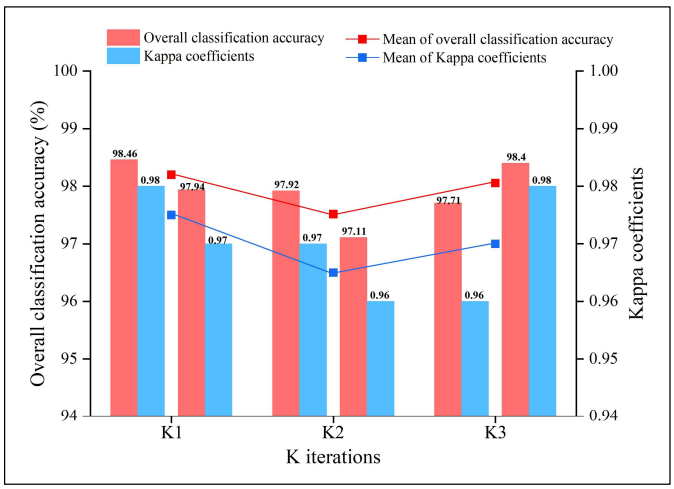


Fig. 15. Overall classification accuracy and Kappa coefficient of three iterations ($K1$–$K3$) from confusion matrix calculation.

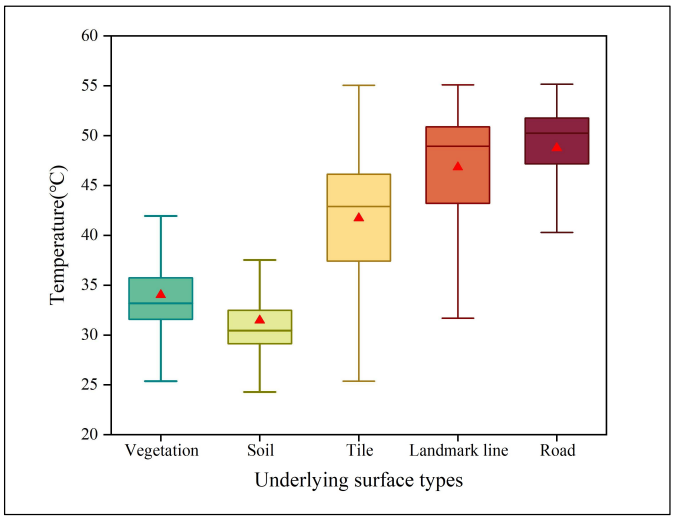


Fig. 16. Box plots of LST for road, landmark line, tile, vegetation, and soil.

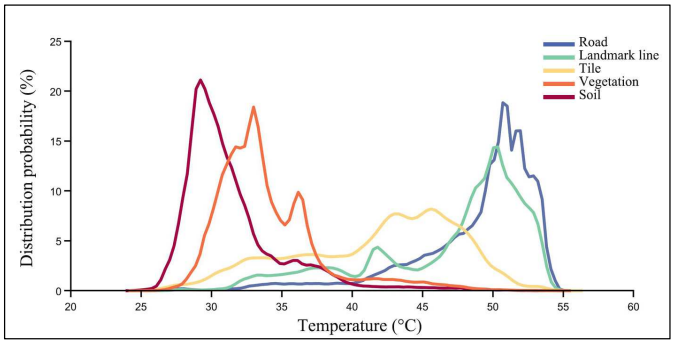


Fig. 17. Temperature distribution probability for road, landmark line, tile, vegetation, and soil underlying surfaces.

demonstrates that the platform could provide stable angle control during the data acquisition. Combined with the image quality analysis result (no image compression and distortion), the LADSRSP exhibits great functionality and performance to acquire continuous and high-precision remote sensing data in low-altitude aerial photography. It is important to note that the attitude variations of the platform during the transitional stages significantly influence IMU values. To avoid this influence, the IMU data were filtered before the analysis according to the time of the transitional stages observed in the thermal infrared image.

B. Assessment Metrics for Image Classification Accuracy

The K-fold cross validation and the confusion matrix were used to evaluate the accuracy of hyperspectral image classification results. In the K-fold cross validation, the number of iterations depends on the K value. The research of Marcot and Hanea [47] suggests that the classification error decreases with the increase of the K value. However, a K value of 5 is sufficient even in the case of large samples. In our study, the test data cover a relatively small area of approximately 0.038 km^2. To avoid repeated selection of the feature regions during the creation of classification samples, the K value was set at 3.

Furthermore, the Kappa coefficient is a statistical measure used to evaluate the agreement between the classification results and the validation samples, ranging between −1 and 1 [48]. Here, a value of 1 indicates total agreement, while a negative value suggests that the agreement is lower than random chance [48]. Overall classification accuracy was often used to assess classification performance, reflecting the percentage of correctly classified samples out of the total; U.S. Geological Survey (USGS) sets a standard of 85% interpretive accuracy for land use and land cover classifications [49], [50]. In our accuracy evaluation, the results demonstrate a Kappa coefficient of 0.97 and an overall classification accuracy of 97.92%. These results demonstrate that the category of pixels can be accurately identified from the hyperspectral data collected by the platform during image classification.

C. Analysis of LST Distribution

In the LST analysis results, the temperature distributions of road, vegetation, soil, and landmark line were leptokurtic (kurtosis > 3; explanation [51]), and the temperature distribution of tile was platykurtic (kurtosis < 3). Furthermore, the temperature distributions of road, tile, and landmark line were positively skewed (skewness > 0; explanation [52]), and the temperature distributions of vegetation and soil were negatively skewed (skewness < 0). In this study, the average temperature of the landmark line was 46.83 °C, whereas that of the road was 48.76 °C. The temperature difference (1.93 °C) between them is minor. Due to conduction, the road and landmark line were expected to have similar temperatures. The temperature difference may be explained by the insufficient accuracy of the emissivity correction, resulting from using the uniform emissivity for the observation area. Furthermore, this difference may also be attributed to the influence of surface properties on radiative albedo. Significant amounts of solar radiation are absorbed and stored by asphalt material due to its lower surface radiative albedo [53]. In contrast, landmark lines are typically made from high-albedo materials (e.g., white thermoplastic coatings and epoxy resins), which are able to

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reflect most of the solar radiation [54]. Moreover, the soil and vegetation showed lower average temperatures than the road, tile, and landmark line, which is consistent with the viewpoint that they have significant cooling effects [55], [56]. The moisture in natural surfaces like soil and vegetation helps to dissipate heat through evapotranspiration [57]. Moreover, the average temperature of the soil is lower than that of the vegetation with a difference of 2.55 °C, which may be attributed to the uniform emissivity value for the observation area.

Regarding the data calibrated operation, our hyperspectral and thermal infrared equipment have already been calibrated during manufacture. Atmospheric conditions may have an influence on different levels on the accuracy of thermal infrared data. Particularly, satellite remote sensing may be affected severely due to the longer distance from the ground. Unlike it, our tests were conducted at an altitude of around 100 m, where the radiance differences between the underlying surface and the sensors are minor [58]. Thus, atmospheric calibration during the tests had little influence on the results. In future studies, the calibration of the hyperspectral sensor will be carried out regularly using the radiometric calibration plate, and the thermal infrared data calibration could be conducted using the simultaneous ground temperature tests on different underlying surfaces to ensure the accuracy of the retrieval of LST.

Furthermore, the thermal infrared equipment has been calibrated during manufacture in two modes (high gain and low gain). As our thermal infrared data were collected in the high gain mode, LST is calculated by transforming the radiance values into temperature values using (1) provided by Teledyne FLIR. Since the atmospheric conditions have little influence on the results at the flight altitude [58], the equation provided by the equipment manufacturer was directly applied to conduct the LST calculation.

It should be noted that different urban underlying surfaces, such as blocks and trees, may have different emissivity values. However, only one emissivity parameter is allowed to be set in the sensor setting process. Moreover, the emissivity parameter cannot be adjusted during aerial observation. Therefore, we cannot set different emissivity parameters for different underlying surfaces. Nevertheless, based on ground test results, emissivity parameters could be adjusted in the data processing software to correct temperature results. In future studies, the thermal infrared data could be acquired with no emissivity correction (epsilon = 1) and then be corrected during postprocessing to conduct a more accurate inversion of LST.

D. Innovation and Comparison Analysis

Compared with UAV-based hyperspectral platforms, the major advantage of our LADSRSP is that it integrates both hyperspectral and thermal infrared sensors to synchronize the acquisition of two different types of remote sensing data (thermal infrared and hyperspectral). UAV-based hyperspectral platforms commonly acquire only one type of hyperspectral data out of the thermal infrared band. Our LADSRSP could synchronously collect thermal infrared and hyperspectral data with high spatial resolution, allowing accurate underlying surface classification and in-depth thermal environment analysis of cities.

This platform has the unique advantage during usage in urban landscapes as it has several distinct characteristics, including rapid deployment capability, synchronous acquisition of dual-source data, and high spatial resolution. First, with rapid deployment capability in complex urban environments and less limitations in site conditions for landing and taking off, this platform is suitable for quick and efficient observation of urban landscapes. Second, two types of remote sensing data with high spatial resolution could be acquired by this platform synchronously. Thermal infrared data can identify the spatial differentiation of surface heat effects, providing key support for optimizing the layout of cooling urban landscapes to mitigate the UHI effect [59]. Hyperspectral data with detailed spectral information could support accurate classification of urban functional zones and provide more precise landscape indicators for urban landscape patterns analysis [60]. Besides, synchronized data could ensure the data homogeneity (i.e., the consistency of data collected under similar or identical environmental conditions) [61], [62]. This is particularly important in urban landscapes, where synchronized data acquisition may avoid the impact of changes in the climatic environment. Finally, the dual-source remote sensing data with high spatial resolution could be used for accurate classification of urban landscapes to support more in-depth and accurate urban landscape studies. Furthermore, the synchronously collected dual-source remote sensing data of LADSRSP enhanced the reliability and accuracy of data collaborative processing, enabling precise identification of urban heat sources and the evaluation of the impact of urban characteristics on LST [63].

In previous studies, a variety of power systems have been proposed and applied to different remote sensing tasks, including fixed-wing drones [64], inflatable uncrewed airships [65], and rigid uncrewed airships made of carbon/epoxy composite [66]. However, the operation of a fixed-wing drone was complex and difficult due to its reliance on a runway for takeoff and landing, and the hovering capability of this drone was limited due to the design of the fixed-wing drone. Besides, uncrewed airships have the limitations of poor portability and low stability due to their large size, heavy weight, and weak wind resistance. In contrast, the multirotor drone was selected as a power system in LADSRSP, which has the portability and flexibility to be quickly deployed in complex urban environments. Specifically, multirotor drones can take off and land vertically without relying on runways and have hovering capability. Moreover, its foldable quad-rotor wings could allow quick assembly, simplify the carrying and transportation process, and provide superior wind resistance.

Two types of platforms have been mainly adopted in previous UAV-based studies on urban thermal environments. One type is the commercial platform with single-source sensors (e.g., thermal infrared or multispectral sensor), which is widely used as it is easy to obtain and does not need to be developed. For instance, Fabbri and Costanzo [67] calibrated an outdoor microclimate simulation model using an UAV equipped with

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a single thermal infrared sensor. Wang et al. [68] investigated the robust correlation between spectral bands and urban forest species characteristics using an UAV equipped with a single multispectral sensor. The other type is the dual-source remote sensing platform, commonly found in combinations of two types of sensors among thermal infrared sensors, visible camera, multispectral sensor, and LiDAR [37], [69], [70]. However, multispectral sensors and visible cameras have difficulty in recognizing minor variations in the underlying urban surface due to their limits in spectral number and spectral resolution, whereas LiDAR does not provide thermal or spectral information and is therefore not suitable for direct thermal environment analysis.

In contrast, the sensors used in our LADSRSP have advantages in function and performance compared to those commonly used in previous research. First, compared with the multispectral sensors commonly equipped on UAVs, the hyperspectral sensor could collect more spectral information between 400 and 1000 nm, with a spatial resolution of 1.47 nm [69]. Moreover, the thermal infrared sensors are more compact and lighter than common handheld thermal infrared cameras while offering a high integration in control and data storage functionalities [71]. Besides, the sensors used in LADSRSP are all nonimaging sensors, which reduces the internal imaging components such as display screens and image processing units commonly installed in imaging sensors. This design not only enhances the portability of sensors but also reduces the power consumption required for operation, making it more suitable for low-altitude drone applications.

E. Environmental Significance, Application Prospects, and Future Work

The LADSRSP provides an innovative tool with high spatial resolution and high spectral resolution for studying the urban thermal environment. Previous studies have utilized satellite remote sensing data (covering visible, near-infrared, and thermal infrared bands) with spatial resolutions between 30 and 1000 m, to analyze urban thermal environment and the relationships with its impact factors (e.g., blue–green infrastructure cooling effects and urban form parameters) [72], [73]. However, Rejaul [72] found that LST satellite data with a spatial resolution of 100 m may significantly affect the analysis results of the cooling effect of blue–green infrastructures and therefore resampled the data with a spatial resolution of 30 m in their study. Furthermore, Bai et al. [74] pointed out that land cover data with 30-m spatial resolution inevitably overlooks smaller green spaces in identifying urban green spaces, and thus, higher spatial resolution data are necessary for distinguishing the changes of urban green spaces in complex urban environments. Although high-precision remote sensing data could be obtained from UAVs, research by Shao et al. [75] indicates that visible images with limited bands may adversely impact the accuracy of image classification. However, hyperspectral images obtained by UAVs have been proven to be effective in identifying vegetation species based on their spectral characteristic differences [76]. Anna et al. [77] suggested that hyperspectral data have a positive effect in mitigating the errors in image identification and classification.

In contrast, the LADSRSP could collect thermal infrared data (9.3 cm) and hyperspectral data (9.4 cm) synchronously with high spatial resolution at low altitude, improving data homogeneity and observational accuracy. The rich spectral data, combined with high-precision thermal infrared data, facilitate the accurate classification of the urban underlying surface and the corresponding temperature analysis. Therefore, our study provides a feasible, efficient, and accurate remote sensing tool with an innovative method, improving the accuracy, comprehensiveness, in-depth, and effectiveness of studies on urban thermal environments, urban greening, and surface temperature analysis in the future.

In this study, the observational area during the tests was insufficient for large-scale urban thermal environment research. Given this, future studies should be conducted in broader regions and complex urban environments to validate the workflow and performance of the LADSRSP in large-scale monitoring. Moreover, future observation experiments could be conducted on a scene with tall buildings to test the image distortion and assess how the walls are captured. Furthermore, as our research mainly focused on the platform itself rather than the analysis of the thermal environment, the longwave radiation and shortwave radiation have not been collected during data acquisition. In future studies, more environmental meteorological parameters (i.e., solar radiation, air temperature, relative humidity, wind speed and direction, and precipitation) could be acquired to enrich the dataset during the observation experiment. In addition, in situ comparative observational experiments are needed to assess its application for in-depth analyses of urban thermal environments.

VI. Conclusion

In our study, a portable and effective LADSRSP for urban thermal environment observation was researched and developed through a series of stability tests, data acquisition tests, and collected data quality analysis tests. A three-axis stabilized gimbal and an IPM were designed and manufactured to integrate and optimize LADSRSP based on these tests. Our results indicate that LADSRSP has good aerial photography performance, with a standard deviation of less than 1° in the control of the roll and pitch axes (roll: 0.524° and pitch: 0.914°). Furthermore, the dual-source data synchronously collected by LADSRSP achieve a spatial resolution of 9.3 cm for thermal infrared data and 9.4 cm for hyperspectral data after resampling. Moreover, the classification results for five urban underlying surface types (road, vegetation, soil, tile, and landmark line) achieved high accuracy with an overall classification accuracy of 97.92% and a Kappa coefficient of 0.97. Finally, the LST of five urban underlying surface types was extracted based on the data fusion between the high-precision classification results and Celsius temperature mosaic, revealing the temperature distribution of permeable (road, landmark line, and tile) and impermeable (vegetation and soil) urban underlying surfaces. In general, through aerial photography tests and the analysis of collected data, the LADSRSP was proven to have several advantages in the field of urban thermal environment, including high stability, high integration, high data quality, and good portability. These characteristics

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could enable reliable dual-source data collection in urban environments and ensure accurate analysis of thermal infrared and hyperspectral data, contributing to in-depth research on urban climate, urban resilience and sustainability, and urban thermal environment.

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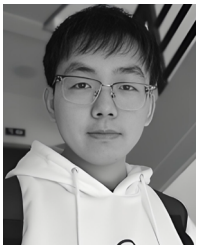
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